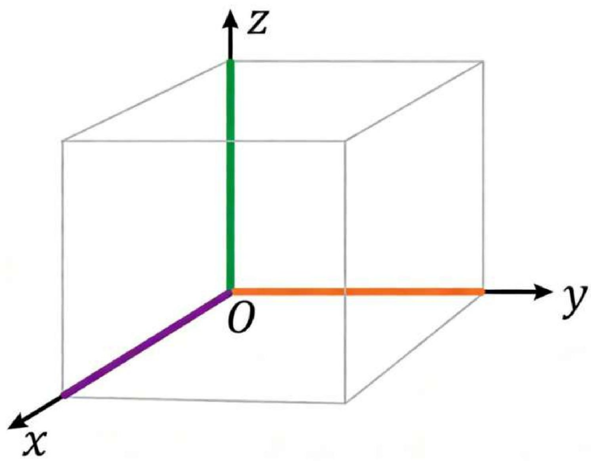
空间直角坐标系综合习题（中英双语版）

一、基础练习 (Basic Exercises)

第 1 题 (Question 1)：在空间直角坐标系中，尝试标出下列各点的位置：(In the space rectangular coordinate system, try to mark the positions of the following points: $A(0,2,4)$, $B(1,0,5)$ 、 $C(0,2,0)$ 、 $D(1,3,4)$ 。

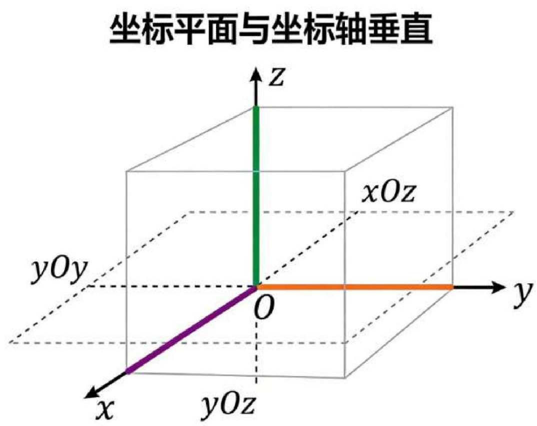


答案 (Answer)：各点位置标注如下：点 A 在 y 轴（x=0）上，y=2、z=4 处；点 B 在 x=1、y=0（xOz 平面）、z=5 处；点 C 在 y 轴（x=0）上，y=2、z=0（xOy 平面）处；点 D 在 x=1、y=3、z=4 处。(The positions of each point are marked as follows: Point A is on the y-axis (x=0) at y=2 and z=4; Point B is at x=1, y=0 (xOz plane) and z=5; Point C is on the y-axis (x=0) at y=2 and z=0 (xOy plane); Point D is at x=1, y=3 and z=4.)

答案解析 (Answer Analysis)：空间直角坐标系中，点的坐标 (x,y,z) 对应三维空间中的位置：x 值表示点到 yOz 平面的距离，y 值表示点到 xOz 平面的距离，z 值表示点到 xOy 平面的距离。标注时，先根据 x、y 值确定在对应坐标平面的投影位置，再沿 z 轴方向平移至对应 z 值处即可。例如点 A(0,2,4)，x=0 说明在 yOz

平面内，先找到 y=2、z=0 的点，再向上平移 4 个单位长度即为 A 点位置。(In the space rectangular coordinate system, the coordinate (x, y, z) of a point corresponds to its position in 3D space: the x-value represents the distance from the point to the yOz plane, the y-value represents the distance to the xOz plane, and the z-value represents the distance to the xOy plane. When marking, first determine the projection position on the corresponding coordinate plane according to the x and y values, then translate along the z-axis to the corresponding z-value. For example, point A(0,2,4), x=0 indicates it is in the yOz plane; first find the point with y=2 and z=0, then translate upward by 4 unit lengths to get the position of point A.)

第 2 题 (Question 2)：在空间直角坐标系 $Oxyz$ 中，哪个坐标平面与 x 轴垂直？哪个坐标平面与 y 轴垂直？哪个坐标平面与 z 轴垂直？(In the space rectangular coordinate system $Oxyz$, which coordinate plane is perpendicular to the x-axis? Which coordinate plane is perpendicular to the y-axis? Which coordinate plane is perpendicular to the z-axis?)

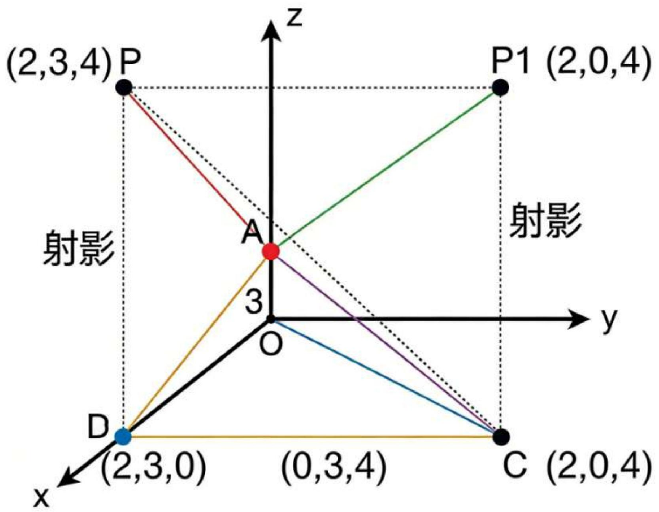


答案 (Answer)：yOz 平面（x=0）与 x 轴垂直；xOz 平面（y=0）与 y 轴垂直；

xOy 平面（z=0）与 z 轴垂直。(The yOz plane (x=0) is perpendicular to the x-axis; the xOz plane (y=0) is perpendicular to the y-axis; the xOy plane (z=0) is perpendicular to the z-axis.)

答案解析 (Answer Analysis)：坐标平面的定义是由两条坐标轴确定的平面，其方程特征为某一坐标值恒为 0。坐标轴与坐标平面垂直的判定依据：若坐标轴的方向向量与平面的法向量平行，则坐标轴与平面垂直。x 轴的方向向量为(1,0,0)，yOz 平面（x=0）的法向量为(1,0,0)，二者平行，故 x 轴与 yOz 平面垂直；同理，y 轴方向向量(0,1,0)与 xOz 平面（y=0）法向量(0,1,0)平行，z 轴方向向量(0,0,1)与 xOy 平面（z=0）法向量(0,0,1)平行，因此对应平面分别与各轴垂直。(A coordinate plane is defined as a plane determined by two coordinate axes, and its equation is characterized by a constant coordinate value of 0. The criterion for a coordinate axis to be perpendicular to a coordinate plane is: if the direction vector of the coordinate axis is parallel to the normal vector of the plane, then the coordinate axis is perpendicular to the plane. The direction vector of the x-axis is (1,0,0), and the normal vector of the yOz plane (x=0) is (1,0,0), which are parallel, so the x-axis is perpendicular to the yOz plane; similarly, the direction vector of the y-axis (0,1,0) is parallel to the normal vector of the xOz plane (y=0) (0,1,0), and the direction vector of the z-axis (0,0,1) is parallel to the normal vector of the xOy plane (z=0) (0,0,1), so the corresponding planes are perpendicular to each axis respectively.)

第 3 题 (Question 3)： 写出点 $P(2,3,4)$ 在三个坐标平面内的射影的坐标。(Write the coordinates of the projection of point $P(2,3,4)$ in the three coordinate planes.)



答案 (Answer)： (www.sicyes.com) 在 xOy 平面内的射影坐标为(2,3,0)；在 yOz 平面内的射影坐标为(0,3,4)；在 xOz 平面内的射影坐标为(2,0,4)。(The projection coordinate in the xOy plane is (2,3,0); the projection coordinate in the yOz plane is (0,3,4); the projection coordinate in the xOz plane is (2,0,4).)

答案解析 (Answer Analysis)： 点在坐标平面内的射影，本质是过该点作坐标平面的垂线，垂足即为射影点。射影点的坐标特征：在哪个坐标平面内射影，对应的那个平面缺失的坐标值恒为 0。具体来说，xOy 平面的方程是 z=0，因此点在 xOy 平面的射影 z 坐标为 0，x、y 坐标保持不变；yOz 平面的方程是 x=0，射影 x 坐标为 0，y、z 坐标不变；xOz 平面的方程是 y=0，射影 y 坐标为 0，x、z 坐标不变。结合点 P(2,3,4), 即可得出三个射影点的坐标。(The projection of a point in a coordinate plane is essentially drawing a perpendicular line from the point to the coordinate plane, and the foot of the perpendicular is the projection point.

The coordinate feature of the projection point: when projecting in a certain coordinate plane, the coordinate value missing from that plane is always 0. Specifically, the equation of the xOy plane is z=0, so the z-coordinate of the point's projection on the xOy plane is 0, and the x and y coordinates remain unchanged; the equation of the yOz plane is x=0, the x-coordinate of the projection is 0, and the y and z coordinates remain unchanged; the equation of the xOz plane is y=0, the y-coordinate of the projection is 0, and the x and z coordinates remain unchanged. Combined with point P(2,3,4), the coordinates of the three projection points can be obtained.)

第 4 题 (Question 4)： 写出点 $P(1,3,5)$ 关于原点成中心对称的点的坐标。(Write the coordinates of the point that is centrally symmetric to point $P(1,3,5)$ with respect to the origin.)

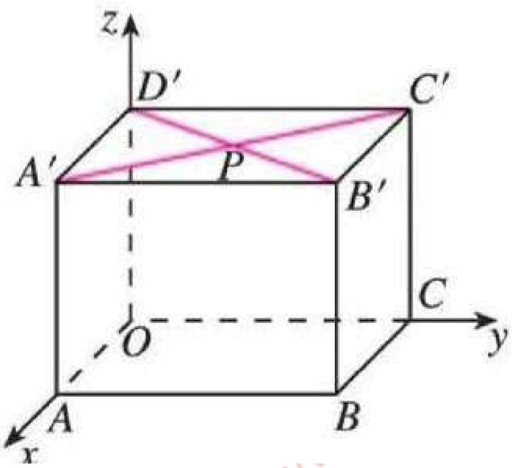
答案 (Answer)： 关于原点成中心对称的点的坐标为(-1,-3,-5)。(The coordinate of the point that is centrally symmetric with respect to the origin is (-1,-3,-5).)

答案解析 (Answer Analysis)： 空间中两点关于原点中心对称的性质：对称点的横、纵、竖坐标均与原点点的对应坐标互为相反数。即若原点点坐标为 (x,y,z) ，则其关于原点的中心对称点坐标为 $(-x,-y,-z)$ 。本题中原点点 P(1,3,5)，因此对称点的 x 坐标为-1，y 坐标为-3，z 坐标为-5，即坐标为(-1,-3,-5)。(The property of central symmetry of two points in space with respect to the origin: the horizontal, the abscissa, ordinate and z-coordinate of the symmetric point are opposite to the corresponding coordinates of the original point. That is, if the coordinate of the original point is (x, y, z) , then the coordinate of its centrally symmetric point with respect to the origin is $(-x, -y, -z)$. In this question, the original point P is (1,3,5), so the x-coordinate of the symmetric point is -1, the y-coordinate is -3, and the z-coordinate is -5, that is, the coordinate is (-1,-3,-5).)

第 5 题 (Question 5)：在长方体 $OABC-D'A'B'C'$ 中， $OA=3$ ， $OC=4$ ， $OD'=3$ ， $A'C'$ 与 $B'D'$ 相交于点 P ，建立空间直角坐标系 $Oxyz$ 。

5.1 写出点 C 、 B' 、 P 的坐标。(In the cuboid $OABC-D'A'B'C'$, $OA=3$, $OC=4$, $OD'=3$, and $A'C'$ intersects $B'D'$ at point P . Establish a space rectangular coordinate system $Oxyz$ and write the coordinates of points C 、 B' 、 P .)

答案 (Answer)：点 C 的坐标为(0,4,0)；点 B’的坐标为(3,4,3)；点 P 的坐标为(1.5,2,3)。(The coordinate of point C is (0,4,0); the coordinate of point B’ is (3,4,3); the coordinate of point P is (1.5,2,3).)



答案解析 (Answer Analysis)：建立空间直角坐标系时，以 O 为原点，OA 所在直线为 x 轴，OC 所在直线为 y 轴，OD’所在直线为 z 轴。根据长方体的顶点特征：①点 C 在 y 轴上，OA=0、OC=4、OD’=0，故坐标为(0,4,0)；②点 B’的横坐标与 OA 相等（3），纵坐标与 OC 相等（4），竖坐标与 OD’相等（3），故坐标为(3,4,3)；③长方体上底面 A’B’C’D’为矩形，矩形的对角线互相平分，因此 P 为 A’C’的中点。先求 A’坐标为(3,0,3)，C’坐标为(0,4,3)，

根据中点坐标公式 $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},\frac{z_1+z_2}{2}\right)$ ，可得 P 点坐标为 $\left(\frac{3+0}{2},\frac{0+4}{2},\frac{3+3}{2}\right)=(1.5,2,3)$ 。(When establishing the space rectangular coordinate system, take O as the origin, the line where OA is located as the x-axis, the line where OC is located as the y-axis, and the line where OD' is located as the z-axis. According to the vertex characteristics of the cuboid: ① Point C is on the y-axis, OA=0, OC=4, OD'=0, so the coordinate is (0,4,0); ② The abscissa of point B' is equal to OA (3), the ordinate is equal to OC (4), and the vertical coordinate is equal to OD' (3), so the coordinate is (3,4,3); ③ The upper base A'B'C'D' of the cuboid is a rectangle, and the diagonals of the rectangle bisect each other, so P is the midpoint of A'C'. First, find the coordinate of A' as (3,0,3) and the coordinate of C' as (0,4,3). According to the midpoint coordinate formula $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},\frac{z_1+z_2}{2}\right)$, the coordinate of point P can be obtained as $\left(\frac{3+0}{2},\frac{0+4}{2},\frac{3+3}{2}\right)=(1.5,2,3)$.)

5.2 写出向量 $\overrightarrow{BB'}$ 、 $\vec{A}'C'$ 的坐标。(Following Question 5, write the coordinates of the vectors $\overrightarrow{BB'}$ and $\vec{A}'C'$.)

答案 (Answer)：向量 $\overrightarrow{BB'}$ 的坐标为(0,0,3); 向量 $\overrightarrow{A'C'}$ 的坐标为(-3,4,0)。(The coordinate of vector $\overrightarrow{BB'}$ is (0,0,3); the coordinate of vector $\overrightarrow{A'C'}$ is (-3,4,0).)

答案解析 (Answer Analysis)：空间向量的坐标求解规则：若向量的起点坐标为 (x_1,y_1,z_1) ，终点坐标为 (x_2,y_2,z_2) ，则向量的坐标为 $(x_2-x_1,y_2-y_1,z_2-z_1)$ 。①由第 5 题可知，B 点坐标为(3,4,0)，B'点坐标为(3,4,3)，因此 $\overrightarrow{BB'}$ 的坐标为(3-3,4-4,3-0)=(0,0,3)；②A'点坐标为(3,0,3)，C'点坐标为(0,4,3)，因此 $\overrightarrow{A'C'}$ 的坐标为(0-3,4-0,3-3)=(-3,4,0)。(The rule for solving the coordinates of a space vector: if the coordinate of the starting point of the vector is (x_1,y_1,z_1) and the coordinate of the ending point is (x_2, y_2, z_2) , then the coordinate of the vector is $(x_2-x_1, y_2-y_1, z_2-z_1)$. ① From Question 5, we know that the coordinate of point B is (3,4,0) and the coordinate of point B' is (3,4,3), so the coordinate of $\overrightarrow{BB'}$ is (3-3,4-4,3-0)=(0,0,3); ② The coordinate of point A' is (3,0,3) and the coordinate of point C' is (0,4,3), so the coordinate of $\overrightarrow{A'C'}$ is (0-3,4-0,3-3)=(-3,4,0).)

第 6 题 (Question 6)： 已知点 B 是点 $A(3,4,5)$ 在坐标平面 Oxy 内的射影，求 $|\overrightarrow{OB}|$ （即线段 OB 的长度）。(It is known that point B is the projection of point $A(3,4,5)$ in the coordinate plane Oxy . Find $|\overrightarrow{OB}|$ (i.e., the length of line segment OB).)

答案 (Answer)： $|\overrightarrow{OB}| = 5$ 。

答案解析 (Answer Analysis)： 第一步，先求射影点 B 的坐标。根据射影坐标特征，点 A 在 xOy 平面的射影 B 的 z 坐标为 0，x、y 坐标与 A 相同，因此 B 点坐标为(3,4,0)。第二步，求向量 $\overrightarrow{OB}$ 的长度。(www.sicyes.com) 空间中，若点 M 的坐标为 (x, y, z) ，则向量 $\overrightarrow{OM}$ 的长度（即原点到点 M 的距离）公式为 $|\overrightarrow{OM}| = \sqrt{x^2+y^2+z^2}$ 。代入 B 点坐标(3,4,0)，可得 $|\overrightarrow{OB}| = \sqrt{3^2+4^2+0^2} = \sqrt{9+16} = \sqrt{25} = 5$ 。(Step 1: First find the coordinate of the projection point B. According to the characteristics of projection coordinates, the z-coordinate of the projection B of point A on the xOy plane is 0, and the x and y coordinates are the same as those of A, so the coordinate of point B is (3,4,0). Step 2: Find the length of vector $\overrightarrow{OB}$. In space, if the coordinate of point M is (x, y, z) , then the length of vector $\overrightarrow{OM}$ (i.e., the distance from the origin to point M) is given by the formula $|\overrightarrow{OM}| =$

$\sqrt{x^2+y^2+z^2}$. Substituting the coordinate of point B (3,4,0), we get $|\overrightarrow{OB}| = \sqrt{3^2+4^2+0^2} = \sqrt{9+16} = \sqrt{25} = 5$.)

二、选择题 (Multiple Choice Questions)

第 1 题 (Question 1): 下列关于右手直角坐标系的判断方法，正确的是（ ）

(Which of the following is correct about the judgment method of the right-handed rectangular coordinate system?)

A. 右手拇指指向 y 轴正方向，食指指向 x 轴正方向，中指指向 z 轴正方向（The thumb of the right hand points to the positive direction of the y -axis, the index finger points to the positive direction of the x -axis, and the middle finger points to the positive direction of the z -axis）

B. 右手拇指指向 x 轴正方向，食指指向 z 轴正方向，中指指向 y 轴正方向（The thumb of the right hand points to the positive direction of the x -axis, the index finger points to the positive direction of the z -axis, and the middle finger points to the positive direction of the y -axis）

C. 右手拇指指向 x 轴正方向，食指指向 y 轴正方向，中指指向 z 轴正方向（The thumb of the right hand points to the positive direction of the x -axis, the index finger points to the positive direction of the y -axis, and the middle finger points to the positive direction of the z -axis）

D. 右手拇指指向 z 轴正方向，食指指向 y 轴正方向，中指指向 x 轴正方向（The thumb of the right hand points to the positive direction of the z -axis, the index finger points to the positive direction of the y -axis, and the middle finger points to the positive direction of the x -axis）

答案（Answer）： C

答案解析（Answer Analysis）： 右手直角坐标系的核心判断规则：右手拇指指向 x 轴正方向，食指指向 y 轴正方向，此时中指自然弯曲所指的方向即为 z 轴正方向。对照选项，只有选项 C 符合该规则；A 选项混淆了 x 轴和 y 轴的指向；B 选项混淆了 y 轴和 z 轴的指向；D 选项三个坐标轴的指向均错误。(The core judgment rule of the right-handed rectangular coordinate system: the thumb of the right hand points to the positive direction of the x-axis, the index finger points to the positive direction of the y-axis, and the direction naturally bent by the middle finger at this time is the positive direction of the z-axis. Comparing the options, only option C conforms to this rule; option A confuses the directions of the x-axis and y-axis; option B confuses the directions of the y-axis and z-axis; option D has incorrect directions for all three coordinate axes.)

第 2 题（Question 2）： 已知空间中两点 $M(2,-1,3)$ 和 $N(4,3,-1)$，则向量 $\overrightarrow{MN}$ 的坐标为（ ）(Given two points $M(2,-1,3)$ and $N(4,3,-1)$ in space, the coordinate of vector $\overrightarrow{MN}$ is ( ))

A. $(2,4,-4)$ B. $(-2,-4,4)$ C. $(6,2,2)$ D. $(1,1,1)$

答案（Answer）： A

答案解析（Answer Analysis）： 向量坐标求解公式为：若向量起点为 $M(x_1,y_1,z_1)$，终点为 $N(x_2,y_2,z_2)$，则 $\overrightarrow{MN}=(x_2-x_1,y_2-y_1,z_2-z_1)$。代入 M(2,-1,3)和 N(4,3,-1)，计算得：x 坐标 4-2=2，y 坐标 3-(-1)=4，z 坐标-1-3=-4，因此 $\overrightarrow{MN}=(2,4,-4)$，对应选项 A。选项 B 是 $\overrightarrow{NM}$ 的坐标，C、D 选项为错误计算结果。(The formula for solving the vector coordinate is: if the starting point of the vector is $M(x_1,y_1,z_1)$ and the ending point is $N(x_2,y_2,z_2)$, then $\overrightarrow{MN}=(x_2-x_1,y_2-y_1,z_2-z_1)$. Substituting M(2,-1,3) and N(4,3,-1), we calculate: x-coordinate 4-2=2, y-coordinate 3-(-1)=4, z-coordinate -1-3=-4, so $\overrightarrow{MN}=(2,4,-4)$, corresponding to option A. Option B is the coordinate of $\overrightarrow{NM}$, and options C and D are incorrect calculation results.)

第 3 题 (Question 3)：空间直角坐标系中， Oyz 平面是由哪两条坐标轴确定的？ （ ）(In the space rectangular coordinate system, which two coordinate axes determine the Oyz plane? ( ))

A. x 轴和 y 轴（ x -axis and y -axis）

B. y 轴和 z 轴（ y -axis and z -axis）

C. x 轴和 z 轴（ x -axis and z -axis）

D. 以上都不对（None of the above）

答案 (Answer)：B

答案解析 (Answer Analysis)：坐标平面的命名规则为：平面名称由其包含的两条坐标轴决定，且以原点 O 为前缀。Oyz 平面的名称中包含 y 和 z，说明该平面由 y 轴和 z 轴相交确定，其平面方程为 x=0（所有点的 x 坐标均为 0）。选项 A 对应的是 xOy 平面，选项 C 对应的是 xOz 平面，因此正确答案为 B。(The naming rule of the coordinate plane is: the plane name is determined by the two coordinate axes it contains, with the origin O as the prefix. The name of the Oyz plane contains y and z, indicating that the plane is determined by the intersection of the y-axis and z-axis, and its plane equation is x=0 (the x-coordinate of all points is 0). Option A corresponds to the xOy plane, option C corresponds to the xOz plane, so the correct answer is B.)

第 4 题 (Question 4)：关于空间（www.sicyes.com）中点 $P(x,y,z)$ 的坐标几何意义，下列说法正确的是（ ）(Regarding the geometric significance of the coordinates of point $P(x,y,z)$ in space, which of the following is correct? ( ))

A. x 是点 P 在 y 轴上的射影坐标（ x is the projection coordinate of point P on the y -axis）

B. y 是点 P 在 z 轴上的射影坐标（ y is the projection coordinate of point P on the z -axis）

C. z 是点 P 在 z 轴上的射影坐标（ z is the projection coordinate of point P on the z -axis）

D. 坐标 (x,y,z) 与向量 $\overrightarrow{OP}$ 的坐标无关（The coordinate (x,y,z) has nothing to do with the coordinate of vector $\overrightarrow{OP}$ ）

答案 (Answer)：C

答案解析 (Answer Analysis)：逐一分析选项：①A 选项错误，点 P 在 y 轴上的射影坐标应为 y，而非 x；②B 选项错误，点 P 在 z 轴上的射影坐标应为 z，而非 y；③C 选项正确，点 P 在 z 轴上的射影是过 P 作 z 轴的垂线，垂足的坐标为(0,0,z)，因此 z 是其在 z 轴上的射影坐标；④D 选项错误，向量 $\overrightarrow{OP}$ 的起点为原点 O(0,0,0)，终点为 P(x,y,z)，根据向量坐标公式，其坐标与 P 点坐标相同，即 $\overrightarrow{OP}=(x,y,z)$ ，二者密切相关。综上，正确答案为 C。(Analyze each option one by one: ① Option A is wrong; the projection coordinate of point P on the y-axis should be y, not x; ② Option B is wrong; the projection coordinate of point P on the z-axis should be z, not y; ③ Option C is correct; the projection of point P on the z-axis is to draw a perpendicular line from P to the z-axis, and the coordinate of the foot of the perpendicular is (0,0,z), so z is its projection coordinate on the z-axis; ④ Option D is wrong; the starting point of vector $\overrightarrow{OP}$ is the origin O(0,0,0), and the ending point is P(x,y,z). According to the vector coordinate formula, its coordinate is the same as the coordinate of point P, that is, $\overrightarrow{OP}=(x,y,z)$, which are closely related. In summary, the correct answer is C.) > （www. sicyes. com）